Point inversion or Central inversion

In geometry, a **point reflection** (also called a **point inversion** or **central inversion**) is a geometric transformation of affine space in which every point is reflected across a designated **inversion center**, which remains fixed. In Euclidean or pseudo-Euclidean spaces, a point reflection is an isometry (preserves distance).1(https://en.wikipedia.org/wiki/Point_reflection#cite_note-1) In the Euclidean plane, a point reflection is the same as a half-turn rotation (180° or π radians), while in three-dimensional Euclidean space a point reflection is an improper rotation which preserves distances but reverses orientation. A point reflection is an involution: applying it twice is the identity transformation.

An object that is invariant under a point reflection is said to possess **point symmetry** (also called **inversion symmetry** or **central symmetry**). A point group including a point reflection among its symmetries is called *centrosymmetric*. Inversion symmetry is found in many crystal structures and molecules, and has a major effect upon their physical properties.2(https://en.wikipedia.org/wiki/Point_reflection#cite_note-:0-2)

Terminology

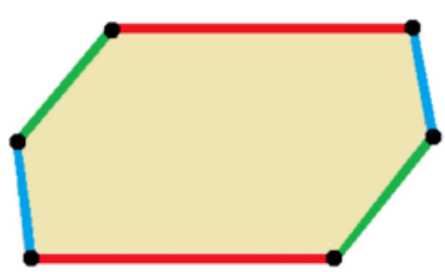
The term *reflection* is loose, and considered by some an abuse of language, with *inversion* preferred; however, *point reflection* is widely used. Such maps are involutions, meaning that they have order 2 – they are their own inverse: applying them twice yields the identity map – which is also true of other maps called *reflections*. More narrowly, a *[reflection](https://en.wikipedia.org/wiki/Reflection*(linear*algebra) "Reflection (linear algebra)")* refers to a reflection in a hyperplane (n−1$n-1$ dimensional affine subspace – a point on the line, a line in the plane, a plane in 3-space), with the hyperplane being fixed, but more broadly *reflection* is applied to any involution of Euclidean space, and the fixed set (an affine space of dimension *k*, where 1≤k≤n−1$1 \leq k \leq n-1$) is called the *mirror*. In dimension 1 these coincide, as a point is a hyperplane in the line.

In terms of linear algebra, assuming the origin is fixed, involutions are exactly the diagonalizable maps with all eigenvalues either 1 or −1. Reflection in a hyperplane has a single −1 eigenvalue (and multiplicity n−1$n-1$ on the 1 eigenvalue), while point reflection has only the −1 eigenvalue (with multiplicity *n*).

The term *inversion* should not be confused with inversive geometry, where *inversion* is defined with respect to a circle.

Examples

2D examples



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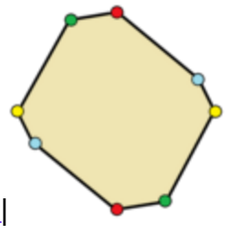


Hexagonal parallelogon|
Octagon|

In two dimensions, a point reflection is the same as a rotation of 180 degrees. In three dimensions, a point reflection can be described as a 180-degree rotation composed with reflection across the plane of rotation, perpendicular to the axis of rotation. In dimension *n*, point reflections are orientation-preserving if *n* is even, and orientation-reversing if *n* is odd.

Formula

Given a vector **a** in the Euclidean space **R***n*, the formula for the reflection of **a** across the point **p** is

Refp(a)=2p−a.$\mathrm{Ref}_{\mathbf{p}}(\mathbf{a}) = 2\mathbf{p} - \mathbf{a}.$

In the case where **p** is the origin, point reflection is simply the negation of the vector **a**.

In Euclidean geometry, the **inversion** of a point *X* with respect to a point *P* is a point *X such that P is the midpoint of the line segment with endpoints X and X*. In other words, the vector from *X* to *P* is the same as the vector from *P* to *X**.

The formula for the inversion in *P* is

x* = 2**p** − **x**

where **p**, **x** and **x*** are the position vectors of *P*, *X* and *X** respectively.

This mapping is an isometric involutive affine transformation which has exactly one fixed point, which is *P*.

Point reflection as a special case of uniform scaling or homothety

When the inversion point *P* coincides with the origin, point reflection is equivalent to a special case of uniform scaling: uniform scaling with scale factor equal to −1. This is an example of linear transformation.

When *P* does not coincide with the origin, point reflection is equivalent to a special case of homothetic transformation: homothety with homothetic center coinciding with P, and scale factor −1. (This is an example of non-linear affine transformation.)

Point reflection group

The composition of two offset point reflections in 2-dimensions is a translation.

The composition of two point reflections is a translation.3(https://en.wikipedia.org/wiki/Point_reflection#cite_note-3) Specifically, point reflection at **p** followed by point reflection at **q** is translation by the vector 2(**q** − **p**).

The set consisting of all point reflections and translations is Lie subgroup of the Euclidean group. It is a semidirect product of **R***n* with a cyclic group of order 2, the latter acting on **R***n* by negation. It is precisely the subgroup of the Euclidean group that fixes the line at infinity pointwise.

In the case *n* = 1, the point reflection group is the full isometry group of the line.

Point reflections in mathematics

- Point reflection across the center of a sphere yields the antipodal map.
- A symmetric space is a Riemannian manifold with an isometric reflection across each point. Symmetric spaces play an important role in the study of Lie groups and Riemannian geometry.

Point reflection in analytic geometry

Given the point P(x,y)$P(x,y)$ and its reflection P′(x′,y′)$P'(x',y')$ with respect to the point C(xc,yc)$C(x_c,y_c)$, the latter is the midpoint of the segment PP′$\overline{PP'}$;

{xc=x+x′2yc=y+y′2 $\begin{cases} x_c = \frac{x+x'}{2} \\ y_c = \frac{y+y'}{2} \end{cases}$

Hence, the equations to find the coordinates of the reflected point are

{x′=2xc−xy′=2yc−y $\begin{cases} x' = 2x_c - x \\ y' = 2y_c - y \end{cases}$

Particular is the case in which the point C has coordinates (0,0)$(0,0)$ (see the paragraph below)

{x′=−xy′=−y $\begin{cases} x' = -x \\ y' = -y \end{cases}$

Properties

In even-dimensional Euclidean space, say 2*N*-dimensional space, the inversion in a point *P* is equivalent to *N* rotations over angles π in each plane of an arbitrary set of *N* mutually orthogonal planes intersecting at *P*. These rotations are mutually commutative. Therefore, inversion in a point in even-dimensional space is an orientation-preserving isometry or direct isometry.

In odd-dimensional Euclidean space, say (2*N* + 1)-dimensional space, it is equivalent to *N* rotations over π in each plane of an arbitrary set of *N* mutually orthogonal planes intersecting at *P*, combined with the reflection in the 2*N*-dimensional subspace spanned by these rotation planes. Therefore, it *reverses* rather than preserves orientation, it is an indirect isometry.

Geometrically in 3D it amounts to rotation about an axis through *P* by an angle of 180°, combined with reflection in the plane through *P* which is perpendicular to the axis; the result does not depend on the orientation (in the other sense) of the axis. Notations for the type of operation, or the type of group it generates, are 1$\bar{1}$, *C__i*, *S_2, and 1×. The group type is one*

of the three symmetry group types in 3D without any pure rotational symmetry, see cyclic symmetries with _n = 1.

The following point groups in three dimensions contain inversion:

- ***Cn_h and _D**n_h for even _n*
- *S_2_n* and *D__n_d for odd _n*
- _T_h, _O_h, and _I_h

Closely related to inverse in a point is reflection in respect to a plane, which can be thought of as an "inversion in a plane".

Inversion centers in crystals and molecules

Inversion symmetry plays a major role in the properties of materials, as also do other symmetry operations.2(https://en.wikipedia.org/wiki/Point_reflection#cite_note-:0-2)

Some molecules contain an inversion center when a point exists through which all atoms can reflect while retaining symmetry. In many cases they can be considered as polyhedra, categorized by their coordination number and bond angles. For example, four-coordinate polyhedra are classified as tetrahedra, while five-coordinate environments can be square pyramidal or trigonal bipyramidal depending on the bonding angles. Six-coordinate octahedra are an example of centrosymmetric polyhedra, as the central atom acts as an inversion center through which the six bonded atoms retain symmetry. Tetrahedra, on the other hand, are non-centrosymmetric as an inversion through the central atom would result in a reversal of the polyhedron. Polyhedra with an odd (versus even) coordination number are not centrosymmtric. Polyhedra containing inversion centers are known as centrosymmetric, while those without are non-centrosymmetric. The presence or absence of an inversion center has a strong influence on the optical properties;4(https://en.wikipedia.org/wiki/Point_reflection#cite_note-4) for instance molecules without inversion symmetry have a dipole moment and can directly interact with photons, while those with inversion have no dipole moment and only interact via Raman scattering.5(https://en.wikipedia.org/wiki/Point_reflection#cite_note-raman1928-5) The later is named after C. V. Raman who was awarded the 1930 Nobel Prize in Physics for his discovery.6(https://en.wikipedia.org/wiki/Point_reflection#cite_note-6)

In addition, in crystallography, the presence of inversion centers for periodic structures distinguishes between centrosymmetric and non-centrosymmetric compounds. All crystalline compounds come from a repetition of an atomic building block known as a unit cell, and these unit cells define which polyhedra form and in what order. In many materials such as oxides these polyhedra can link together via corner-, edge- or face sharing, depending on which atoms share common bonds and also the valence. In other cases such as for metals and alloys the structures are better considered as arrangements of close-packed atoms. Crystals which do not have inversion symmetry also display the piezoelectric effect. The presence or absence of inversion symmetry also has numerous consequences for the properties of solids,2(https://en.wikipedia.org/wiki/Point_reflection#cite_note-:0-2) as does the mathematical relationships between the different crystal symmetries.7(https://en.wikipedia.org/wiki/Point_reflection#cite_note-7)

Real polyhedra in crystals often lack the uniformity anticipated in their bonding geometry. Common irregularities found in crystallography include distortions and disorder. Distortion involves the warping of polyhedra due to nonuniform bonding lengths, often due to differing electrostatic interactions between heteroatoms or electronic effects such as Jahn–Teller distortions. For instance, a titanium center will likely bond evenly to six oxygens in an octahedra, but distortion would occur if one of the oxygens were replaced with a more electronegative fluorine. Distortions will not change the inherent geometry of the

polyhedra—a distorted octahedron is still classified as an octahedron, but strong enough distortions can have an effect on the centrosymmetry of a compound. Disorder involves a split occupancy over two or more sites, in which an atom will occupy one crystallographic position in a certain percentage of polyhedra and the other in the remaining positions. Disorder can influence the centrosymmetry of certain polyhedra as well, depending on whether or not the occupancy is split over an already-present inversion center.

Centrosymmetry applies to the crystal structure as a whole, not just individual polyhedra. Crystals are classified into thirty-two crystallographic point groups which describe how the different polyhedra arrange themselves in space in the bulk structure. Of these thirty-two point groups, eleven are centrosymmetric. The presence of noncentrosymmetric polyhedra does not guarantee that the point group will be the same—two non-centrosymmetric shapes can be oriented in space in a manner which contains an inversion center between the two. Two tetrahedra facing each other can have an inversion center in the middle, because the orientation allows for each atom to have a reflected pair. The inverse is also true, as multiple centrosymmetric polyhedra can be arranged to form a noncentrosymmetric point group.

Inversion with respect to the origin

Inversion with respect to the origin corresponds to additive inversion of the position vector, and also to scalar multiplication by −1. The operation commutes with every other linear transformation, but not with translation: it is in the center of the general linear group. "Inversion" without indicating "in a point", "in a line" or "in a plane", means this inversion; in physics 3-dimensional reflection through the origin is also called a parity transformation.

In mathematics, **reflection through the origin** refers to the point reflection of Euclidean space **R***n* across the origin of the Cartesian coordinate system. Reflection through the origin is an orthogonal transformation corresponding to scalar multiplication by −1-1, and can also be written as −I$-I$, where II is the identity matrix. In three dimensions, this sends (x,y,z)↦(−x,−y,−z)$(x, y, z) \mapsto (-x, -y, -z)$, and so forth.

Representations

As a scalar matrix, it is represented in every basis by a matrix with −1-1 on the diagonal, and, together with the identity, is the center of the orthogonal group O(n)$O(n)$.

It is a product of *n* orthogonal reflections (reflection through the axes of any orthogonal basis); note that orthogonal reflections commute.

In 2 dimensions, it is in fact rotation by 180 degrees, and in dimension 2n$2n$, it is rotation by 180 degrees in *n* orthogonal planes;a(https://en.wikipedia.org/wiki/Point_reflection#cite_note-8) note again that rotations in orthogonal planes commute.

Properties

It has determinant (−1)n$(-1)^n$ (from the representation by a matrix or as a product of reflections). Thus it is orientation-preserving in even dimension, thus an element of the special orthogonal group SO(2*n*), and it is orientation-reversing in odd dimension, thus not an element of SO(2*n* + 1) and instead providing a splitting of the map O(2n+1)→±1$O(2n+1) \to \pm 1$, showing that O(2n+1)=SO(2n+1)×{±I}$O(2n+1) = SO(2n+1) \times \{\pm I\}$ as an internal direct product.

- Together with the identity, it forms the center of the orthogonal group.
- It preserves every quadratic form, meaning Q(−v)=Q(v)$Q(-v) = Q(v)$, and thus is an element of every indefinite orthogonal group as well.

- It equals the identity if and only if the characteristic is 2.
- It is the longest element of the Coxeter group of signed permutations.

Analogously, it is a longest element of the orthogonal group, with respect to the generating set of reflections: elements of the orthogonal group all have length at most *n* with respect to the generating set of reflections,b(https://en.wikipedia.org/wiki/Point_reflection#cite_note-9) and reflection through the origin has length *n,* though it is not unique in this: other maximal combinations of rotations (and possibly reflections) also have maximal length.

Geometry

In SO(2*r*), reflection through the origin is the farthest point from the identity element with respect to the usual metric. In O(2*r* + 1), reflection through the origin is not in SO(2*r*+1) (it is in the non-identity component), and there is no natural sense in which it is a "farther point" than any other point in the non-identity component, but it does provide a base point in the other component.

Clifford algebras and spin groups

Further information: Clifford algebra

Further information: Spin group

It should *not* be confused with the element −1∈Spin(n)$-1 \in \mathrm{Spin}(n)$ in the spin group. This is particularly confusing for even spin groups, as −I∈SO(2n)$-I \in SO(2n)$, and thus in Spin(n) $\mathrm{Spin}(n)$ there is both −1-1 and 2 lifts of −I$-I$.

Reflection through the identity extends to an automorphism of a Clifford algebra, called the *main involution* or *grade involution.*

Reflection through the identity lifts to a pseudoscalar.

Geometrical optics

Geometrical optics, or **ray optics**, is a model of optics that describes light propagation in terms of *[rays](https://en.wikipedia.org/wiki/Ray*(optics) "Ray (optics)")_. The ray in geometrical optics is an abstraction useful for approximating the paths along which light propagates under certain circumstances.

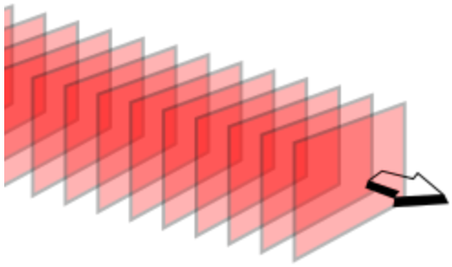
The simplifying assumptions of geometrical optics include that light rays:

- propagate in straight-line paths as they travel in a homogeneous medium
- bend, and in particular circumstances may split in two, at the interface between two dissimilar media
- follow curved paths in a medium in which the refractive index changes
- may be absorbed or reflected.

Geometrical optics does not account for certain optical effects such as diffraction and interference, which are considered in physical optics. This simplification is useful in practice; it is an excellent approximation when the wavelength is small compared to the size of structures with which the light interacts. The techniques are particularly useful in describing geometrical aspects of imaging, including optical aberrations.

Explanation

Further information: Ray (optics) and Ray tracing (physics)



As light travels through space, it oscillates in amplitude. In this image, each maximum amplitude crest is marked with a plane to illustrate the wavefront. The ray is the arrow perpendicular to these parallel surfaces.

A light ray is a line or curve that is perpendicular to the light's wavefronts (and is therefore collinear with the wave vector). A slightly more rigorous definition of a light ray follows from Fermat's principle, which states that the path taken between two points by a ray of light is the path that can be traversed in the least time.1(https://en.wikipedia.org/wiki/Geometrical_optics#cite_note-1)

Geometrical optics is often simplified by making the paraxial approximation, or "small angle approximation". The mathematical behavior then becomes linear, allowing optical components and systems to be described by simple matrices. This leads to the techniques of Gaussian optics and *paraxial [ray tracing](https://en.wikipedia.org/wiki/Ray_tracing*(physics) "Ray tracing (physics)")_, which are used to find basic properties of optical systems, such as approximate image and object positions and magnifications.2(https://en.wikipedia.org/wiki/Geometrical_optics#cite_note-Greivenkamp-2)

Reflection

Main article: Reflection (physics)

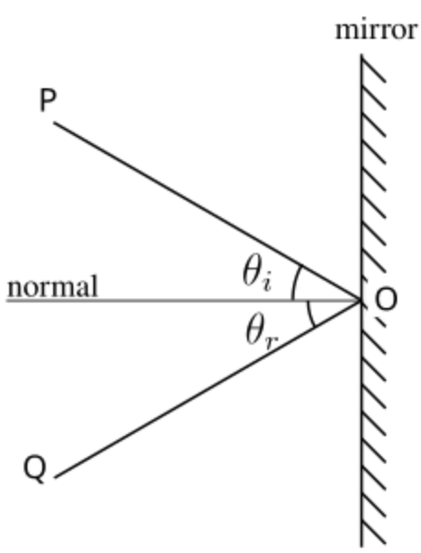


Diagram of specular reflection

Glossy surfaces such as mirrors reflect light in a simple, predictable way. This allows for production of reflected images that can be associated with an actual (real) or extrapolated (virtual) location in space.

With such surfaces, the direction of the reflected ray is determined by the angle the incident ray makes with the surface normal, a line perpendicular to the surface at the point where the ray hits. The incident and reflected rays lie in a single plane, and the angle between the reflected ray and the surface normal is the same as that between the incident ray and the normal.3(https://en.wikipedia.org/wiki/Geometrical_optics#cite_note-Geoptics-3) This is known as the Law of Reflection.

For flat mirrors, the law of reflection implies that images of objects are upright and the same distance behind the mirror as the objects are in front of the mirror. The image size is the same as the object size. (The magnification of a flat mirror is equal to one.) The law also implies that mirror images are parity inverted, which is perceived as a left-right inversion.

Mirrors with curved surfaces can be modeled by ray tracing and using the law of reflection at each point on the surface. For mirrors with parabolic surfaces, parallel rays incident on the mirror produce reflected rays that converge at a common focus. Other curved surfaces may also focus light, but with aberrations due to the diverging shape causing the focus to be smeared out in space. In particular, spherical mirrors exhibit spherical aberration. Curved mirrors can form images with magnification greater than or less than one, and the image can be upright or inverted. An upright image formed by reflection in a mirror is always virtual, while an inverted image is real and can be projected onto a screen.3(https://en.wikipedia.org/wiki/Geometrical_optics#cite_note-Geoptics-3)

Refraction

	This section **should include a summary of, or be summarized in, another article.** See Wikipedia:Summary style for information on how to incorporate it into this article's main text, or the main text of another article. *(June 2009)*

Main article: Refraction

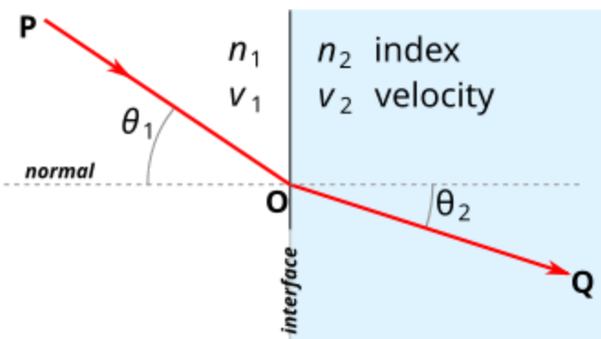


Illustration of Snell's Law

Refraction occurs when light travels through an area of space that has a changing index of refraction. The simplest case of refraction occurs when there is an interface between a uniform medium with index of refraction n1n_1 and another medium with index of refraction n2n_2. In such situations, Snell's Law describes the resulting deflection of the light ray:n1sinθ1=n2sinθ2 $n_1 \sin\theta_1 = n_2 \sin\theta_2$where θ1$\theta_1$ and θ2θ_2 are the angles between the normal (to the interface) and the incident and refracted waves, respectively. This phenomenon is also associated with a changing speed of light as seen from the definition of index of refraction provided above which implies:v1sinθ2 =v2sinθ1$v_1 \sin\theta_2 \ = v_2 \sin\theta_1$where v1$v_1$ and v2v_2 are the wave velocities through the respective media.3(https://en.wikipedia.org/wiki/Geometrical_optics#cite_note-Geoptics-3)
Because of the physical considerations one can assume without loss of generality that φφ is of the following form: φ(x,y,z,t)=ψ(x,y,z)−ct$\varphi(x,y,z,t) = \psi(x,y,z) - ct$, i.e. a 2D surface moving through space, modelled as level surfaces of ψψ. (Mathematically ψψ exists if φt≠0$\varphi_t \neq 0$ by the implicit function theorem.) The above equation written in terms of ψψ becomes:‖∇ψ‖2=εμc2(−c)2=εμ=n2$\|\nabla\psi\|^2 = \frac{\varepsilon\mu}{c^2}(-c)^2 = \varepsilon\mu = n^2$ i.e.,ψx2+ψy2+ψz2=n2 $\psi_x^2 + \psi_y^2 + \psi_z^2 = n^2$which is the eikonal equation and it holds for all xx, yy, zz, since the variable tt is absent. Other laws of optics like Snell's law and Fresnel formulae can be similarly obtained by considering discontinuities in εε and μμ.

General equation using four-vector notation

In four-vector notation used in special relativity, the wave equation can be written as∂2ψ∂xi∂xi=0 $\frac{\partial^2 \psi}{\partial x_i \partial x^i} = 0$

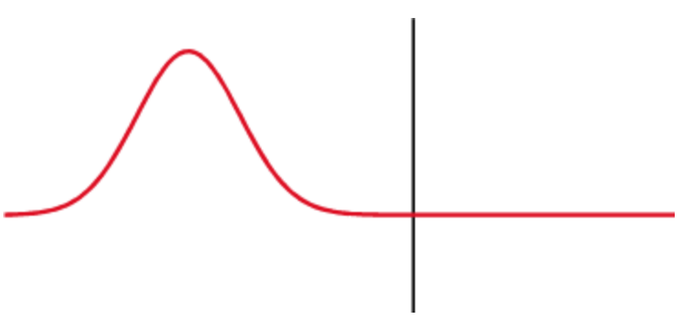
and the substitution ψ=AeiS/ε$\psi = Ae^{iS/\varepsilon}$ leads to12(https://en.wikipedia.org/wiki/Geometrical_optics#cite_note-12)−Aε2∂S∂xi∂S∂xi+2iε∂A∂xi∂S∂xi+iAε∂2S∂xi∂xi+∂2A∂xi∂xi=0.![{\displaystyle -{\frac {A}{\varepsilon ^{2}}}{\frac {\partial S}{\partial x*{i}}}{\frac {\partial S}{\partial x^{i}}}+{\frac {2i}{\varepsilon }}{\frac {\partial A}{\partial x*{i}}}{\frac {\partial S}{\partial x^{i}}}+{\frac {iA}{\varepsilon }}{\frac {\partial ^{2}S}{\partial x*{i}\partial x^{i}}}+{\frac {\partial ^{2}A}{\partial x*{i}\partial x^{i}}}=0.}](https://wikimedia.org/api/rest_v1/media/math/render/svg/234226b235515e1768f833923eeeaa6638498d60)

Therefore, the eikonal equation is given by∂S∂xi∂S∂xi=0. $\frac{\partial S}{\partial x_i}\frac{\partial S}{\partial x^i} = 0.$

Once eikonal is found by solving the above equation, the wave four-vector can be found fromki=−∂S∂xi. $k_i = -\frac{\partial S}{\partial x^i}.$

Reflection coefficient

In physics and electrical engineering the **reflection coefficient** is a parameter that describes how much of a wave is reflected by an impedance discontinuity in the transmission medium. It is equal to the ratio of the amplitude of the reflected wave to the incident wave, with each expressed as phasors. For example, it is used in optics to calculate the amount of light that is reflected from a surface with a different index of refraction, such as a glass surface, or in an electrical transmission line to calculate how much of the electromagnetic wave is reflected by an impedance discontinuity. The reflection coefficient is closely related to the *transmission coefficient*. The reflectance of a system is also sometimes called a reflection coefficient.



A wave is partially transmitted and partially reflected when the medium through which it travels suddenly changes. The reflection coefficient determines the ratio of the reflected wave amplitude to the incident wave amplitude.

Different disciplines have different applications for the term.

Transmission lines

See also: Reflections of signals on conducting lines and Signal reflection

In telecommunications and transmission line theory, the reflection coefficient is the ratio of the complex amplitude of the reflected wave to that of the incident wave. The voltage and

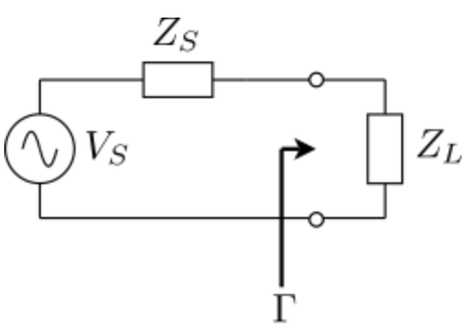
current at any point along a transmission line can always be resolved into forward and reflected traveling waves given a specified reference impedance *Z0*. The reference impedance used is typically the characteristic impedance of a transmission line that's involved, but one can speak of reflection coefficient without any actual transmission line being present. In terms of the forward and reflected waves determined by the voltage and current, the reflection coefficient is defined as the complex ratio of the voltage of the reflected wave (V−V^-) to that of the incident wave (V+V^+). This is typically represented with a ΓΓ (capital gamma) and can be written as:

Γ=V−V+$$\Gamma = \frac{V^-}{V^+}$$

It can also be defined using the *currents* associated with the reflected and forward waves, but introducing a minus sign to account for the opposite orientations of the two currents:

Γ=−I−I+=V−V+$$\Gamma = -\frac{I^-}{I^+} = \frac{V^-}{V^+}$$

The reflection coefficient may also be established using other field or circuit pairs of quantities whose product defines power resolvable into a forward and reverse wave. With electromagnetic plane waves, one uses the ratio of the electric fields of the reflected to that of the incident wave (or magnetic fields, again with a minus sign); the ratio of each wave's electric field *E* to its magnetic field *H* is the medium's characteristic impedance, Z0Z_0, (equal to the impedance of free space if the medium is a vacuum).1(https://en.wikipedia.org/wiki/Reflection_coefficient#cite_note-1)



Simple circuit configuration showing measurement location of reflection coefficient

In the accompanying figure, a signal source with internal impedance ZSZ_S possibly followed by a transmission line of characteristic impedance ZSZ_S is represented by its Thévenin equivalent, driving the load ZLZ_L. For a real (resistive) source impedance ZSZ_S, if we define ΓΓ using the reference impedance Z0=ZS$Z_0 = Z_S$ then the source's maximum power is delivered to a load ZL=Z0$Z_L = Z_0$, in which case Γ=0$\Gamma = 0$ implying no reflected power. More generally, the squared-magnitude of the reflection coefficient |Γ|2$|\Gamma|^2$ denotes the proportion of that power that is reflected back to the source, with the power actually delivered toward the load being 1−|Γ|2$1 - |\Gamma|^2$.

Anywhere along an intervening (lossless) transmission line of characteristic impedance Z0Z_0, the magnitude of the reflection coefficient |Γ|$|\Gamma|$ will remain the same (the powers of the forward and reflected waves stay the same) but with a different phase. In the case of a short circuited load (ZL=0$Z_L = 0$), one finds Γ=−1$\Gamma = -1$ at the load. This implies the reflected wave having a 180° phase shift (phase reversal) with the voltages of the two waves being opposite at that point and adding to zero (as a short circuit demands).

Relation to load impedance

The reflection coefficient is determined by the load impedance at the end of the transmission line, as well as the characteristic impedance of the line. A load impedance of ZLZ_L terminating

a line with a characteristic impedance of Z0Z_0 will have a reflection coefficient of

Γ=ZL−Z0ZL+Z0.$$\Gamma = \frac{Z_L - Z_0}{Z_L + Z_0}.$$

This is the coefficient at the load. The reflection coefficient can also be measured at other points on the line. The *magnitude* of the reflection coefficient in a lossless transmission line is constant along the line (as are the powers in the forward and reflected waves). However its *phase* will be shifted by an amount dependent on the electrical distance ϕϕ from the load. If the coefficient is measured at a point LL meters from the load, so the electrical distance from the load is ϕ=2πL/λ$\phi = 2\pi L/\lambda$ radians, the coefficient Γ′Γ' at that point will be

Γ′=Γe−i2ϕ$\Gamma' = \Gamma e^{-i\,2\phi}$

Note that the phase of the reflection coefficient is changed by *twice* the phase length of the attached transmission line. That is to take into account not only the phase delay of the reflected wave, but the phase shift that had first been applied to the forward wave, with the reflection coefficient being the quotient of these. The reflection coefficient so measured, Γ′Γ', corresponds to an impedance which is generally dissimilar to ZLZ_L present at the far side of the transmission line.

The complex reflection coefficient (in the region |Γ|≤1$|\Gamma| \leq 1$, corresponding to passive loads) may be displayed graphically using a Smith chart. The Smith chart is a polar plot of ΓΓ, therefore the magnitude of ΓΓ is given directly by the distance of a point to the center (with the edge of the Smith chart corresponding to |Γ|=1$|\Gamma| = 1$). Its evolution along a transmission line is likewise described by a rotation of 2ϕ2ϕ around the chart's center. Using the scales on a Smith chart, the resulting impedance (normalized to Z0Z_0) can directly be read. Before the advent of modern electronic computers, the Smith chart was of particular use as a sort of analog computer for this purpose.

The reflected power in terms of the reflection coefficient is:

Preflected=Pincident|Γ|2$P_{reflected} = P_{incident}|\Gamma|^2$.

Standing wave ratio

Main article: Standing wave ratio

The standing wave ratio (SWR) is determined solely by the *magnitude* of the reflection coefficient:

SWR=1+|Γ|1−|Γ|.$$SWR = \frac{1 + |\Gamma|}{1 - |\Gamma|}.$$

Along a lossless transmission line of characteristic impedance *Z_0, the SWR signifies the ratio of the voltage (or current) maxima to minima (or what it would be if the transmission line were long enough to produce them). The above calculation assumes that ΓΓ has been calculated using _Z_0 as the reference impedance. Since it uses only the _magnitude* of ΓΓ, the SWR intentionally ignores the specific value of the load impedance *ZL* responsible for it, but only the magnitude of the resulting impedance mismatch. That SWR remains the same wherever measured along a transmission line (looking towards the load) since the addition of a transmission line length to a load ZLZ_L only changes the phase, not magnitude of ΓΓ. While having a one-to-one correspondence with reflection coefficient, SWR is the most commonly used figure of merit in describing the mismatch affecting a radio antenna or antenna system. It

is most often measured at the transmitter side of a transmission line, but having, as explained, the same value as would be measured at the antenna (load) itself.

Electrical networks

A transmission line is an example of a 2-port electrical network, but reflection coefficients are useful in the analysis of any electrical networks. A reflection coefficient for each port in the same way as for the boundary of a transmission line. It will, however, also depend on the properties of connections at other ports and so is not a property intrinsic to the network itself. For a 2-port network with the 2x2 scattering matrix *S*, and with a source and load connected to its input and output, where the reflections off the source back into the input are ΓSΓ_S and the reflections off the load back into the output are ΓLΓ_L, then the reflection coefficients at the input and output are given by:2(https://en.wikipedia.org/wiki/Reflection_coefficient#cite_note-2)

|Γin|=|S11+S12S21ΓL1−S22ΓL|$|\Gamma_{\text{in}}| = \left| S_{11} + \frac{S_{12}S_{21}\Gamma_L}{1 - S_{22}\Gamma_L} \right|$ and |Γout|=|S22+S12S21ΓS1−S11ΓS|$|\Gamma_{\text{out}}| = \left| S_{22} + \frac{S_{12}S_{21}\Gamma_S}{1 - S_{11}\Gamma_S} \right|$

Seismology

Main article: Reflection seismology

Reflection coefficient is used in feeder testing for reliability of medium.

Optics and microwaves

Main article: Fresnel equations

In optics and electromagnetics in general, *reflection coefficient* can refer to either the amplitude reflection coefficient described here, or the reflectance, depending on context. Typically, the reflectance is represented by a capital *R*, while the amplitude reflection coefficient is represented by a lower-case *r*. These related concepts are covered by Fresnel equations in classical optics.

Acoustics

Main article: Acoustic wave § Reflection

Acousticians use reflection coefficients to understand the effect of different materials on their acoustic environments. The field properties used to define the reflection coefficient are typically the acoustic pressure and velocity in the incident and reflected acoustic waves.